Tutorial 4. Periodic properties of elements

Exercise 1 *(Und)*

Among the elements N, O, P, and S, which element or elements have the smallest effective nuclear charge? Which element or elements have the largest effective nuclear charge?

Exercise 2 *(Know)*

Write the electron configurations for the following ions, and determine which have noble-gas configurations: (**a**) Mg^{2+}, (**b**) Br^{-}, (**c**) Ti^{2+}, (**d**) Po^{2-}, (**e**) V^{3+}

Exercise 3 *(Und)*

Consider the isoelectronic ions F^{-} and Na^{+}. Assuming that core electrons contribute 1.00 and valence electrons contribute 0.00 to the screening constant, *S*, calculate Z_{eff} for the 2*p* electrons in both ions. For isoelectronic ions, how are effective nuclear charge and ionic radius related? Which ion is smaller?

Exercise 4 *(App)*

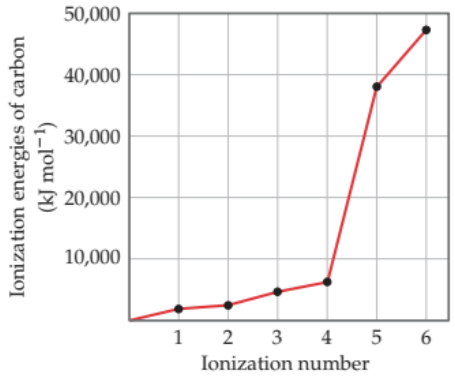
a) Explain the reason for decreasing atomic size from left to right across a period of periodic table.
b) Arrange the atoms and ions in each series in order of decreasing radius:
   (i) Mg^{2+}, Ca^{2+}, and Ca
   (ii) F, S^{2-}, Cl, and Se^{2-}
   (iii) K^{+}, Cl^{-}, Ca^{2+}, and S^{2-}
   (iv) Br^{-}, Rb^{+}, Se^{2-}, Sr^{2+}, and Te^{2-}

Exercise 5 *(App)*

a) Write equations that show the processes that describe the first, second, and third ionization energies of a chlorine atom. Which process would require the least amount of energy?
b) Based on their positions in the periodic table, predict which atom of the following pairs will have the smaller first ionization energy. Explain the answers.
   (i) Br, Kr
   (ii) C, Ca
   (iii) Li, Rb
   (iv) Pb, Si
   (v) Al, B

Exercise 6 *(App)*

Explain the variation in the ionization energies of carbon, as displayed in this graph.



Exercise 7 *(App)*

Which of the following chemical equations is connected to the definitions of (X) the first ionization energy of oxygen, (Y) the second ionization energy of oxygen, and (Z) the electron affinity of oxygen?

(i) $O(g) + e^- \rightarrow O^-(g)$

(ii) $O(g) \rightarrow O^+(g) + e^-$

(iii) $O(g) + 2\ e^- \rightarrow O_2^-(g)$

(iv) $O(g) \rightarrow O_2^+(g) + 2\ e^-$

(v) $O^+(g) \rightarrow O_2^+(g) + e^-$

Exercise 8 *(App)*

(a) Arrange the following series in the order of increasing metallic characteristics:
+ Na, K, Al, Mg, Zn, S, Cl.

(b) Arrange the following series in the order of increasing non-metallic characteristics:
+ Br, F, Cu, Fe, S, C, Ca